



GIS-BASED GRAPE MOTH MONITORING AT QUINTA DA SENHORA DA GRAÇA

BY

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SUBMITTED

TO



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1.0 Background of the Practicals

Quinta da Senhora da Graça is situated near Peso da Régua within the internationally renowned Douro Demarcated Region of northern Portugal, one of the oldest legally protected wine-producing regions in the world. The estate occupies approximately 42.97 hectares (429,690.52 m²) and represents a typical example of the complex vineyard landscapes that characterize the Douro Valley. The practical exercise conducted at this farm was designed to provide students with a comprehensive understanding of the relationship between topography, climate, vineyard management, and pest monitoring within a precision viticulture framework. The Douro region is recognized globally for the production of high-quality wines, including Port wine and premium table wines. The success of viticulture in this region is closely linked to unique environmental conditions, including steep terraced slopes, schist-derived soils, Mediterranean climatic influences, and centuries of accumulated agricultural knowledge. These factors collectively create a distinctive terroir that influences grape development, fruit composition, and wine quality. Modern viticulture increasingly relies on technological innovations to improve productivity while maintaining environmental sustainability. Precision agriculture technologies, particularly Geographic Information Systems (GIS), remote sensing, Global Navigation Satellite Systems (GNSS), and spatial analysis tools, have become essential components of vineyard management. These technologies enable farmers and researchers to monitor spatial variability within vineyards, identify zones of differing productivity, evaluate environmental constraints, and optimize management practices. The practical activities undertaken at Quinta da Senhora da Graça focused on terrain analysis, land-use characterization, varietal distribution assessment, and pest management. Particular emphasis was placed on the grape moth (*Lobesia botrana*), one of the most economically significant vineyard pests in Europe. Through field observations and GIS-based analysis, students were able to examine how topographic variability influences pest distribution, vineyard performance, and management requirements. The practical also highlighted the importance of integrating environmental monitoring with agricultural decision-making. By understanding how slope, altitude, climate, soil characteristics, and pest pressure interact within a vineyard ecosystem, managers can develop more effective and sustainable production strategies. This multidisciplinary approach represents the future of viticulture, where traditional knowledge is combined with advanced technologies to address modern agricultural challenges.

1.1 Geographical Location and Regional Context

Quinta da Senhora da Graça is strategically located on the right bank of the Corgo River, one of the major tributaries of the Douro River. The estate lies within the municipality of Peso da Régua, a city widely regarded as the administrative and commercial center of the Douro wine region. This geographical position places the vineyard at the heart of one of Europe's most important viticultural landscapes. The Douro Valley extends across northeastern Portugal and is characterized by deeply incised river valleys, steep mountainsides, and highly dissected terrain. Over thousands of years, the Douro River and its tributaries have carved extensive valleys through the landscape, creating a complex topography that strongly influences local climatic conditions. These physical characteristics contribute significantly to the uniqueness of the region's wines.

The estate exhibits considerable altitudinal variation, ranging from approximately 61 meters above sea level near the river margins to nearly 400 meters at higher elevations. Such elevation differences create substantial environmental gradients across relatively short distances. Temperature, humidity, solar radiation, wind exposure, and soil moisture conditions vary significantly along these gradients, resulting in the formation of distinct microclimates throughout the property. The proximity of the vineyard to the Corgo River provides several environmental advantages. The river acts as a thermal regulator, moderating temperature extremes during both summer and winter. Water bodies generally absorb and release heat more slowly than surrounding land surfaces, reducing the risk of sudden temperature fluctuations that can negatively affect grape development. Furthermore, the river influences local air circulation patterns, contributing to the movement of cool air through the valley during nighttime periods. However, river proximity may also create favorable conditions for certain diseases and pests. Elevated humidity levels, particularly during specific periods of the growing season, can promote fungal disease development if vineyard management practices are inadequate. Consequently, continuous monitoring and appropriate phytosanitary measures are essential components of vineyard management in such environments. The strategic location of Quinta da Senhora da Graça within the Douro Valley makes it an excellent site for studying the interactions between geography, climate, topography, and agricultural production systems.

1.2 Climate and Weather Characteristics of the Douro Region

The climate of the Douro Valley is classified as Mediterranean with strong continental influences. This climatic regime is characterized by hot, dry summers and relatively cool, wet winters. The unique weather conditions experienced throughout the year play a fundamental role in determining vine growth, grape composition, and overall wine quality. Summer temperatures frequently exceed 35°C during July and August, while temperatures above 40°C are not uncommon in lower valley locations. These high temperatures promote sugar accumulation within grape berries and contribute to the development of characteristic flavor compounds. The long, sunny growing season provides abundant solar radiation, ensuring adequate photosynthetic activity and grape maturation. Winter conditions are considerably cooler, with temperatures occasionally approaching freezing levels in higher elevation areas. Rainfall occurs primarily between October and April, replenishing soil moisture reserves that support vine growth during the subsequent growing season. Annual precipitation generally ranges from 600 to 1,000 mm depending on altitude and exposure. One of the most significant climatic characteristics of the Douro Valley is the presence of strong microclimatic variation. South-facing slopes typically receive greater solar radiation and experience higher temperatures than north-facing slopes. Consequently, grape maturation rates can differ substantially between vineyards located only a short distance apart. The weather conditions experienced during the growing season have direct implications for vineyard management. Prolonged periods of high temperature and limited rainfall may induce water stress, requiring careful irrigation management where permitted. Conversely, excessive humidity during flowering or ripening stages may increase disease risk. Climate change has emerged as a major concern for Douro viticulture. Recent decades have witnessed increasing temperatures, more frequent heatwaves, and changes in precipitation patterns. These climatic shifts may alter grape phenology, increase water requirements, and influence pest and disease dynamics. As a result, precision

monitoring technologies are becoming increasingly important for adapting vineyard management practices to changing environmental conditions.

1.3 Topographic Characteristics and Terrain Morphology

The topography of Quinta da Senhora da Graça represents one of its most defining features. The estate displays remarkable terrain variability, with elevations ranging from 61 m to 400 m and slopes varying from nearly flat areas to extremely steep sections exceeding 111%. Steep slopes are characteristic of the Douro landscape and have historically necessitated the construction of terraces to facilitate agricultural production. These terraces not only enable vine cultivation on otherwise inaccessible terrain but also help reduce soil erosion and improve water retention. The development and maintenance of terraced vineyards represent significant engineering achievements that have shaped the cultural landscape of the region. Topographic variability influences numerous environmental factors that directly affect vine performance. Slope angle determines the amount of solar radiation received by the vineyard surface, while slope orientation influences temperature patterns and moisture availability. South-facing slopes generally experience greater heat accumulation and faster grape maturation, whereas north-facing slopes often maintain cooler conditions and slower ripening rates. Elevation gradients also affect atmospheric conditions. Higher elevations typically experience lower temperatures, greater wind exposure, and different humidity patterns compared to lower valley locations. These environmental differences contribute to spatial variability in vine vigor, fruit quality, and harvest timing. The rugged terrain presents both opportunities and challenges for vineyard management. While topographic diversity contributes to the production of complex and distinctive wines, it also complicates mechanization, increases labor requirements, and necessitates specialized management strategies. Precision agriculture technologies have become increasingly valuable for addressing these challenges by enabling detailed spatial analysis of terrain characteristics. Digital Elevation Models (DEMs) generated through GIS software provide valuable information for assessing slope, aspect, drainage patterns, and erosion risk. Such analyses support informed decision-making regarding vineyard establishment, irrigation planning, soil conservation, and pest management.

1.4 Soils and Geological Characteristics

The soils of Quinta da Senhora da Graça are predominantly derived from schist parent material, a characteristic geological feature of the Douro region. Schist is a metamorphic rock that fractures easily, allowing vine roots to penetrate deeply into the soil profile in search of water and nutrients. These soils are generally shallow, stony, and well-drained. Although such conditions might appear unfavorable for agriculture, they are particularly advantageous for high-quality wine production. Moderate soil fertility limits excessive vegetative growth, encouraging vines to allocate resources toward fruit development rather than leaf production. The fractured nature of schist facilitates deep root penetration, enabling vines to access water reserves during dry summer periods. This characteristic is especially important in the Douro region, where prolonged drought conditions frequently occur during the growing season. Soil properties vary considerably across the estate due to differences in slope position, erosion processes, and weathering intensity. Upper slopes often possess shallower soils with greater stone content, while lower slopes may accumulate

deeper soil deposits through sediment transport. Soil variability significantly influences vine performance and grape quality. Consequently, detailed soil mapping and GIS-based spatial analysis are increasingly employed to identify management zones within vineyards. These approaches allow producers to implement site-specific management practices that optimize productivity while conserving environmental resources.

1.5 Importance of Precision Viticulture and GIS Applications

Precision viticulture represents a modern management approach that utilizes advanced technologies to address spatial variability within vineyards. At Quinta da Senhora da Graça, GIS serves as a fundamental tool for collecting, analyzing, and visualizing spatial information relevant to vineyard management. The adoption of GIS technologies enables vineyard managers to integrate multiple datasets, including topography, soil characteristics, climate information, pest distribution, and crop performance indicators. These integrated analyses provide valuable insights into the factors influencing vineyard productivity and sustainability. One of the principal advantages of GIS is its ability to identify management zones. Different sections of a vineyard often exhibit varying levels of productivity due to differences in environmental conditions. By recognizing these variations, managers can apply inputs such as fertilizers, irrigation water, and pesticides more efficiently. GIS also plays an essential role in environmental monitoring and conservation. Spatial analyses can identify areas vulnerable to erosion, waterlogging, nutrient deficiencies, or pest infestations. Early identification of such problems allows timely interventions that reduce economic losses and environmental impacts. The practical activities conducted at Quinta da Senhora da Graça demonstrated how GIS technologies can support evidence-based decision-making in viticulture. Students gained experience in analyzing spatial datasets and interpreting the relationships between environmental variables and vineyard performance. As climate variability and sustainability concerns continue to influence agricultural systems worldwide, the importance of GIS and precision viticulture is expected to increase further. These technologies provide powerful tools for improving resource-use efficiency, maintaining productivity, and ensuring the long-term viability of viticultural landscapes such as those found in the Douro Valley.

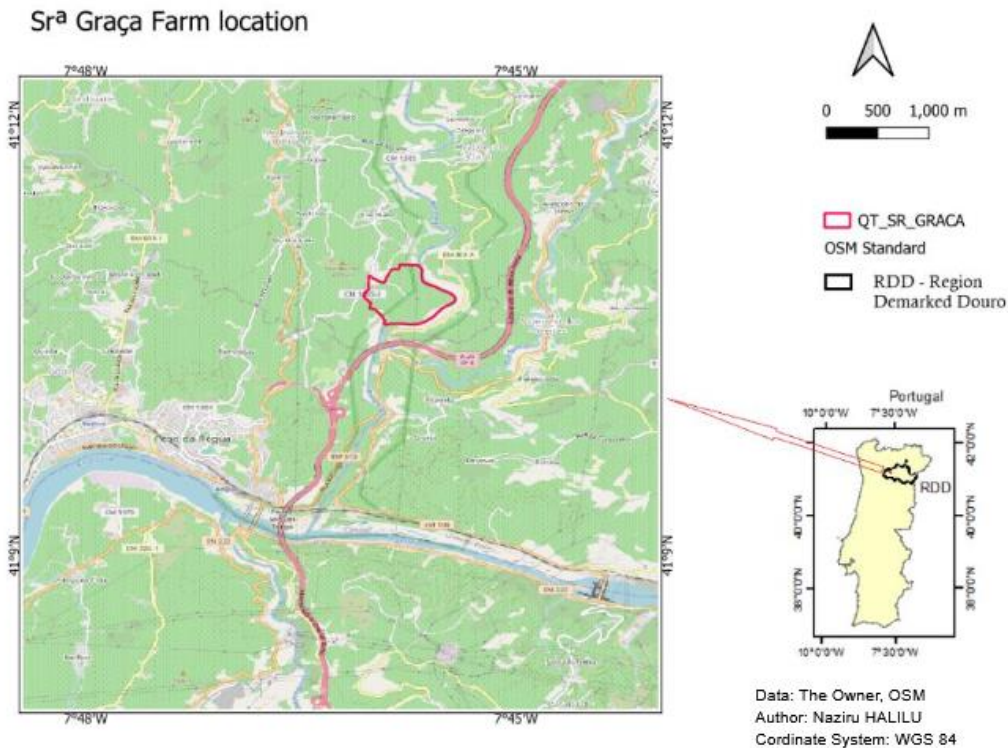


Figure 1 – Geographical and cartographic location of Sr^a da Graça's Qt.

2.0 Altimetric Variation of Quinta da Senhora da Graça

Altimetric variation refers to the differences in elevation that occur across a landscape. At Quinta da Senhora da Graça, the terrain exhibits a significant altitudinal range, with elevations varying from approximately 61 meters above sea level in the areas closest to the Corgo River to about 400 meters above sea level in the upper sections of the property. This elevation difference of nearly 339 meters demonstrates the highly variable relief that characterizes the vineyard and reflects the rugged geomorphology of the Douro Valley. The altimetric variation observed within the estate plays a crucial role in shaping environmental conditions and influencing vineyard productivity. Elevation affects several climatic variables, including temperature, humidity, solar radiation, wind exposure, and rainfall distribution. As altitude increases, temperatures generally decrease, creating cooler environmental conditions in the higher sections of the vineyard. These cooler conditions influence grapevine phenology, delaying budburst, flowering, veraison, and harvest compared to lower-altitude zones. In viticulture, temperature is one of the most important factors affecting grape development. The lower portions of the property experience warmer conditions that promote rapid sugar accumulation and earlier fruit maturation. These areas often produce grapes with higher sugar concentrations and lower acidity. In contrast, the higher sections of the vineyard experience cooler temperatures that slow the ripening process, allowing grapes to retain acidity while developing more complex aromatic and phenolic compounds. This variation contributes

significantly to the diversity of grape characteristics found within the same estate. Altimetric variation also influences the movement of air masses within the vineyard. During the night, cooler and denser air tends to move downslope and accumulate in lower areas, while warmer air remains at higher elevations. This process affects local temperature patterns and can influence frost occurrence, humidity levels, and disease development. Good air circulation is particularly important in vineyards because it reduces the duration of leaf wetness and lowers the risk of fungal diseases. Furthermore, elevation affects water availability and soil moisture dynamics. Higher elevations often experience greater wind exposure and increased evapotranspiration, which may lead to drier conditions during the summer growing season. Lower areas may retain more moisture due to runoff accumulation and proximity to the river. Consequently, different parts of the vineyard may require distinct management strategies to optimize vine performance and fruit quality.

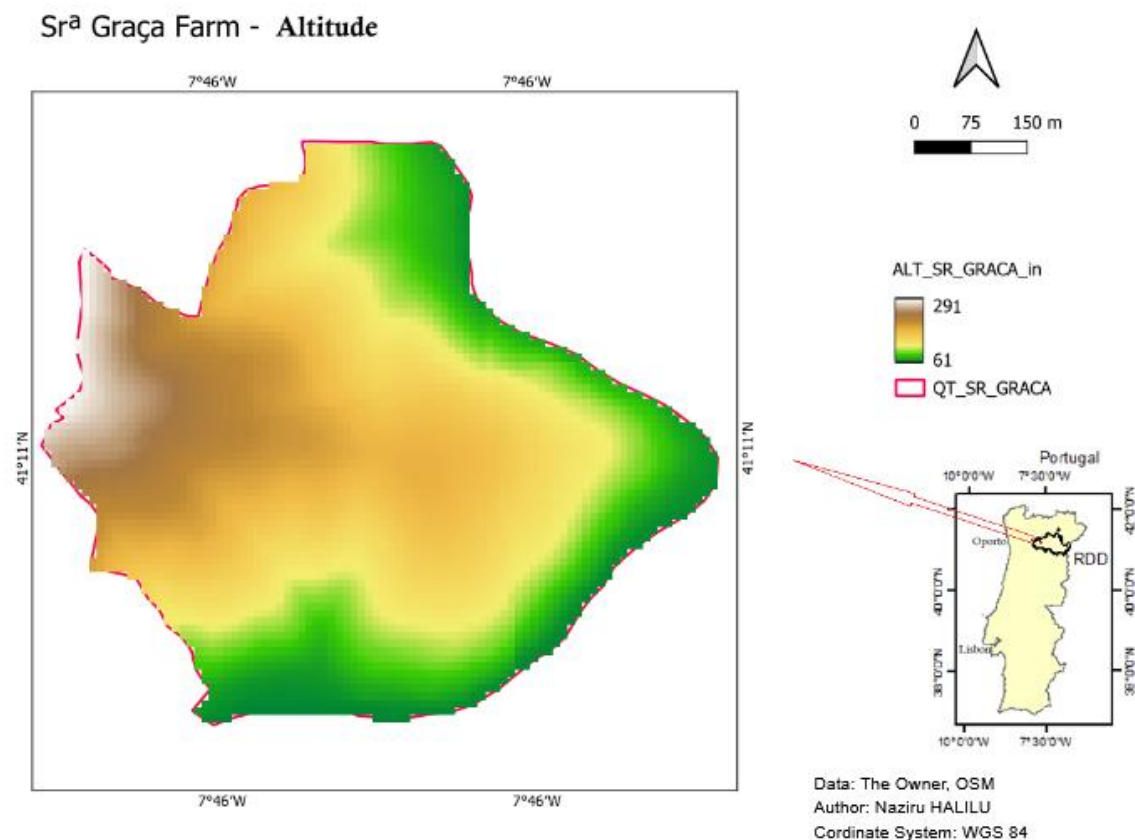


Figure 2- Altitude chart of the Qt.ª da Sr.ª da Graça.

The altimetric diversity of Quinta da Senhora da Graça creates a mosaic of microclimates across the property. These microclimates contribute to variations in vine vigor, grape composition, and harvest timing, making altitude a critical factor in vineyard management. Through the use of Geographic Information Systems (GIS) and Digital Elevation Models (DEMs), vineyard managers can identify and map these variations, enabling the implementation of precision viticulture

practices that improve productivity and sustainability. The practical analysis of altimetric variation demonstrates the importance of topography in shaping agricultural landscapes. Understanding elevation-related differences allows producers to make informed decisions regarding vineyard establishment, irrigation, pest management, and harvesting operations. In the Douro Valley, where terrain complexity is a defining characteristic, altimetric variation is not merely a geographical feature but a key determinant of wine quality and vineyard success.

3.0 Topographic Heterogeneity of the Property

Topographic heterogeneity refers to the spatial variability in terrain characteristics such as elevation, slope, aspect, and landform configuration. At Quinta da Senhora da Graça, topographic heterogeneity is one of the most distinctive features of the landscape. The property occupies approximately 42.97 hectares and displays considerable variation in both elevation and slope, reflecting the rugged terrain that is characteristic of the Douro wine region. The vineyard is situated within a highly dissected valley system formed by the erosive action of the Douro River and its tributaries over geological time. As a result, the landscape consists of steep hillsides, river terraces, ridges, and valleys that create a complex and heterogeneous terrain. Such diversity has profound implications for agricultural production, particularly in viticulture, where environmental conditions strongly influence grapevine growth and fruit quality. One of the most remarkable aspects of the property's topography is the variation in slope gradient. Slope values range from 0% in relatively flat areas to approximately 111% in the steepest sections near the Corgo River. These extreme gradients create significant differences in environmental conditions across the vineyard. Steep slopes receive varying amounts of solar radiation depending on their orientation, influencing temperature regimes and grape maturation patterns. Topographic heterogeneity contributes to the formation of numerous microclimates within the estate. Differences in slope orientation affect the duration and intensity of sunlight exposure. South-facing slopes generally receive greater solar radiation and experience warmer conditions, while north-facing slopes remain cooler and often retain higher moisture levels. These variations influence vine growth rates, grape composition, and the timing of harvest operations. The heterogeneous terrain also affects soil formation and distribution. Upper slopes often possess shallower soils due to erosion processes, whereas lower slopes may accumulate deeper deposits of soil and organic matter. This spatial variability in soil properties influences water-holding capacity, nutrient availability, root development, and vine vigor. Consequently, vineyard productivity can vary considerably from one location to another within the same property. Weather conditions interact strongly with topography. During the hot Mediterranean summers typical of the Douro region, steep slopes and varying elevations create localized differences in temperature and humidity. Areas exposed to intense solar radiation may experience greater heat accumulation and increased evapotranspiration, while shaded zones maintain cooler conditions. Similarly, wind patterns are influenced by terrain configuration, affecting ventilation and disease pressure within the vineyard. Topographic heterogeneity also presents important management challenges. The steep terrain limits the use of conventional

agricultural machinery, requiring specialized equipment or manual labor for many vineyard operations. Activities such as pruning, canopy management, pest monitoring, and harvesting become more labor-intensive and costly on steep slopes. In addition, erosion control measures must be implemented to prevent soil degradation and maintain long-term productivity. Despite these challenges, topographic heterogeneity provides significant advantages for wine production. The diversity of environmental conditions allows the cultivation of different grape varieties under optimal conditions and contributes to the complexity and uniqueness of the wines produced. The variation in microclimates and terroir characteristics enhances the potential for producing grapes with diverse flavor profiles and quality attributes. Modern precision viticulture technologies have become essential tools for managing topographically complex vineyards such as Quinta da Senhora da Graça. Geographic Information Systems (GIS), drone surveys, satellite imagery, and digital terrain analysis enable managers to map and monitor terrain variability with high accuracy. These technologies support site-specific management practices that improve resource-use efficiency, reduce environmental impacts, and enhance grape quality. Overall, the topographic heterogeneity of Quinta da Senhora da Graça represents a defining characteristic of the estate. It influences every aspect of vineyard management, from climate and soil conditions to pest dynamics and harvest planning. Understanding and managing this complexity is fundamental to maintaining the productivity, sustainability, and quality of viticultural production in the Douro Valley.

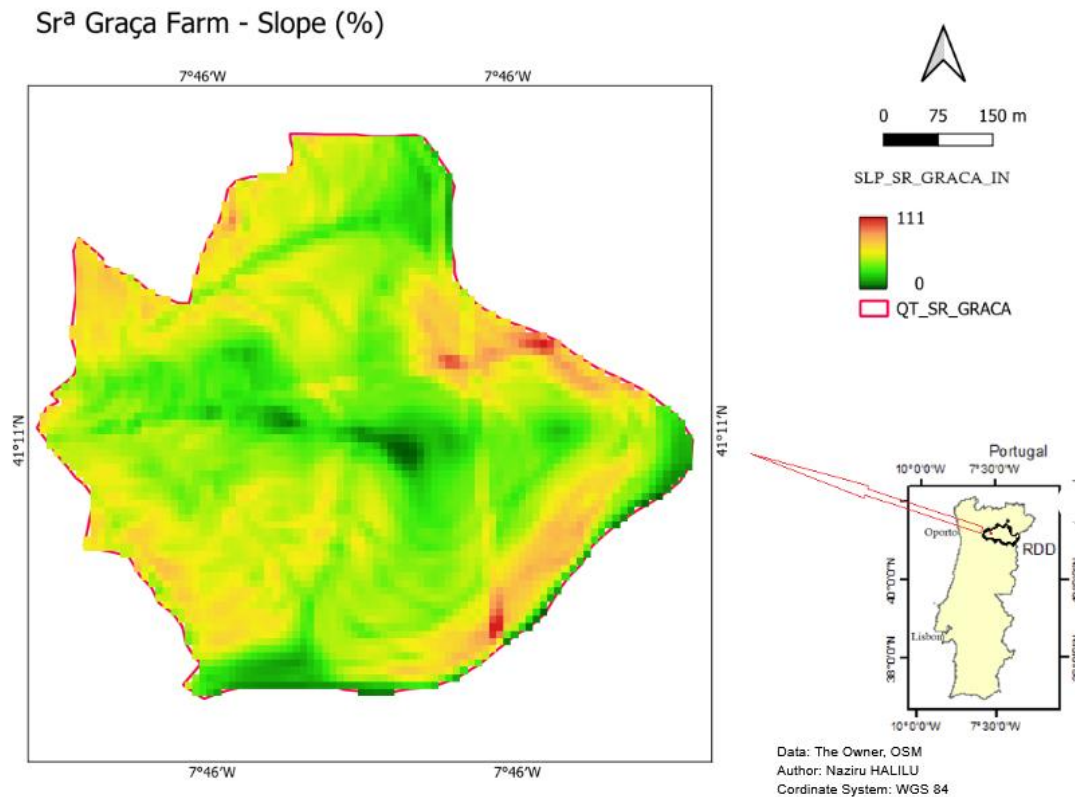


Figure 2- Map of slopes of the Qt.^a da Sr.^a da Graça

4.0 Elevation Profile and Morphological Characteristics of the Property

The elevation profile of Quinta da Senhora da Graça provides important insights into the physical structure of the vineyard and its influence on viticultural production. The estate exhibits considerable morphological heterogeneity, with elevations ranging from approximately 61.69 m above sea level in the lowest areas near the Corgo River to about 290.58 m in the highest sections of the property. This variation in altitude contributes to the formation of diverse environmental conditions that directly affect grapevine growth, grape quality, pest development, and vineyard management practices. The analysis of topographic profiles allows a better understanding of how the terrain influences climatic processes, water movement, and biological interactions within the vineyard ecosystem. The rugged morphology of the property is typical of the Douro Demarcated Region, where centuries of viticulture have adapted to steep slopes and complex terrain conditions. The combination of altitude variation, steep gradients, and diverse slope orientations creates a highly heterogeneous landscape that supports multiple microclimatic zones. Such variability is considered one of the key factors contributing to the uniqueness of Douro wines, as different sections of the vineyard may produce grapes with distinct characteristics despite being located within the same estate. Altitude has a significant effect on local climatic conditions. Higher areas generally experience lower temperatures, increased wind exposure, and greater thermal amplitude

between day and night. These conditions slow grape maturation, allowing for the gradual accumulation of sugars and the preservation of organic acids. Lower areas, particularly those located near the river, tend to be warmer and more humid, promoting faster ripening and increased vegetative growth. The existence of these contrasting environmental conditions within a relatively small area creates opportunities for producing grapes with diverse quality attributes and oenological potential.

The slope distribution within the property ranges from approximately 0.3° to 110.82° , indicating the presence of extremely steep terrain. Such slopes are a defining characteristic of Douro vineyards and significantly influence vineyard operations. Steep slopes enhance solar radiation interception, improve natural drainage, and contribute to favorable conditions for grape cultivation. However, they also present considerable management challenges. Mechanized operations become difficult or impossible in many sections of the vineyard, requiring manual labor for pruning, canopy management, harvesting, and pest control activities. The increased labor requirements contribute to higher production costs but are often justified by the superior grape quality associated with these unique growing conditions. The steep terrain also influences erosion processes and runoff dynamics. During periods of intense rainfall, water moves rapidly downslope, increasing the risk of soil erosion and nutrient loss. Consequently, vineyard managers must implement conservation measures such as terraces, vegetation cover, and drainage systems to minimize environmental degradation and maintain soil productivity. Sustainable soil management is particularly important in the Douro region, where soil depth is often limited and natural regeneration processes are slow. The west-to-east topographic profile of the estate (Figure 4) reveals a general decrease in elevation from the western section toward the eastern portion of the property. This topographic arrangement has several important implications for vineyard management. Water runoff naturally follows this gradient, influencing soil moisture distribution and nutrient transport across the landscape. Areas located in lower positions may receive runoff from higher slopes, potentially increasing soil moisture availability during dry periods. Furthermore, the movement of cold air is influenced by terrain configuration, with cooler air masses tending to descend toward lower elevations during nighttime hours. This process affects local temperature patterns and may influence grape ripening and disease development. The west-to-east profile also plays a role in the dispersion of pheromone clouds used in mating disruption systems for grape moth control. Wind movement and air circulation patterns are strongly influenced by topography, affecting the distribution and effectiveness of pheromone-based pest management strategies. Understanding these spatial relationships is essential for optimizing the placement of pheromone dispensers and ensuring adequate coverage throughout the vineyard. Similarly, the north-to-south profile (Figure 5) indicates the presence of a central elevated zone with descending slopes toward both the northern and southern margins of the property. This configuration contributes to the formation of distinct microclimatic environments within the vineyard. The central elevated area is likely to experience greater wind exposure and lower humidity levels compared to the surrounding lower sections. These conditions may reduce disease

pressure while influencing grapevine water use and physiological responses. The descending slopes facilitate air drainage and contribute to variations in temperature accumulation across different vineyard blocks. The interaction between topography and climate is particularly important in relation to pest management. The grape moth (*Lobesia botrana*), one of the most significant vineyard pests in Europe, is highly influenced by environmental conditions such as temperature, humidity, and air movement. Areas with favorable microclimatic conditions may support higher pest populations, while less favorable zones may experience reduced infestation levels. Understanding the spatial distribution of these conditions enables vineyard managers to implement targeted monitoring and control measures, improving the efficiency of integrated pest management programs. Aspect, or slope orientation, further contributes to the morphological complexity of the estate. The analysis indicates that the property does not possess a dominant orientation but instead presents a heterogeneous pattern of exposure. This diversity creates multiple microclimates characterized by varying levels of sunlight interception, temperature accumulation, and evapotranspiration. South-facing slopes generally receive greater solar radiation and experience warmer conditions, while north-facing slopes tend to remain cooler and retain moisture for longer periods. East-facing slopes benefit from morning sunlight, whereas west-facing slopes are exposed to intense afternoon radiation. According to Aranha (2026), slope orientation influences several key factors affecting grapevine performance, including photosynthetic activity, water use efficiency, canopy development, and fruit maturation. These effects contribute to spatial variability in grape quality and harvest timing across the vineyard. The heterogeneous exposure pattern observed at Quinta da Senhora da Graça therefore represents an important component of the property's terroir, enhancing the diversity and complexity of the grapes produced.

The analysis of the topographic profiles demonstrates that the terrain of Quinta da Senhora da Graça is far from uniform. Instead, it consists of a complex arrangement of elevations, slopes, and aspects that interact to create a mosaic of environmental conditions. This topographic heterogeneity influences virtually every aspect of vineyard management, from irrigation and soil conservation to pest monitoring and harvest planning. Modern precision viticulture technologies, including Geographic Information Systems (GIS), digital terrain modeling, and spatial analysis, provide valuable tools for understanding and managing this complexity. Ultimately, the elevation profiles and morphological characteristics of the property highlight the intimate relationship between landscape structure and viticultural production in the Douro Valley. The unique terrain not only shapes the environmental conditions experienced by the vines but also contributes significantly to the quality, identity, and economic value of the wines produced at Quinta da Senhora da Graça. Through careful analysis and management of these topographic features, vineyard managers can optimize production while preserving the environmental and cultural heritage of this remarkable wine-growing region.

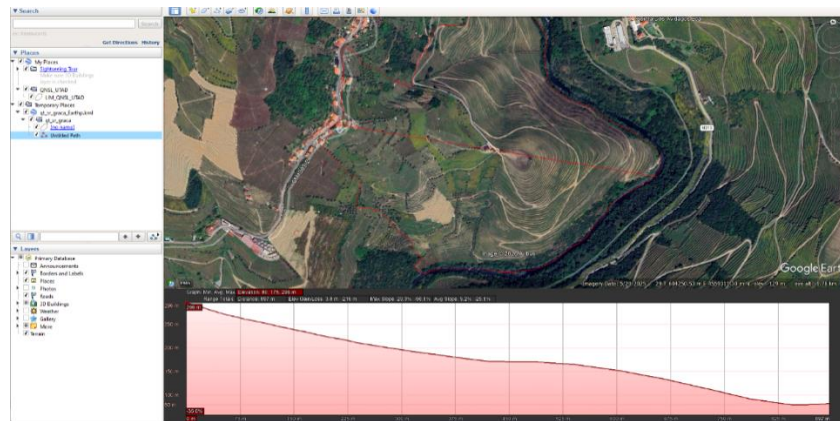


Figure 4 – West to east profile for Sr da Graça farm

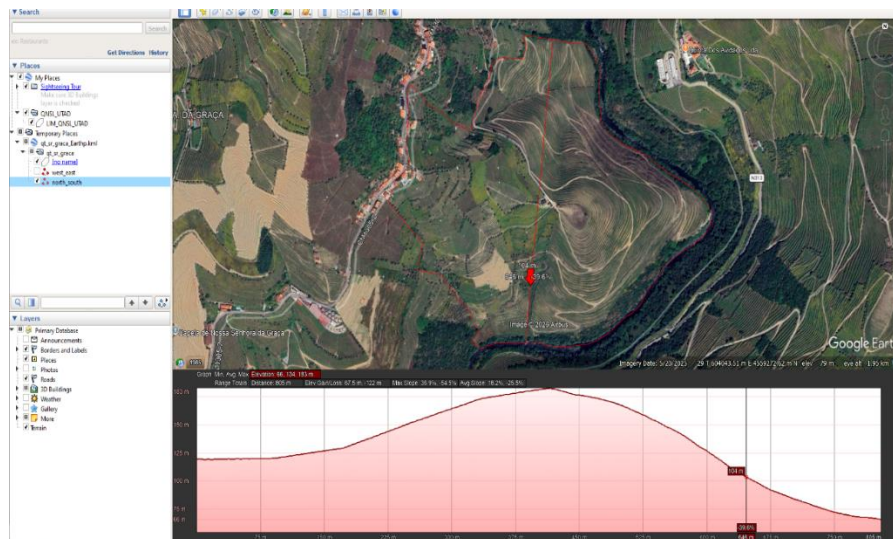


Figure 5 – North to South profile for Sr da Graça farm

5.0 Aspect Model

The aspect model is an important component of terrain analysis that describes the compass direction toward which a slope faces. In viticulture, aspect plays a crucial role because it determines the amount of solar radiation received by different sections of the vineyard, influencing temperature, evapotranspiration, soil moisture, grapevine growth, and fruit maturation. At Quinta da Senhora da Graça, the analysis of the aspect map reveals that the study area does not possess a dominant quadrant. Instead, the property exhibits a heterogeneous distribution of slope orientations (Figure 6), reflecting the complex morphology that characterizes the Douro Valley landscape. This diversity of slope orientations contributes significantly to the environmental heterogeneity of the vineyard. Different aspects receive varying amounts of sunlight throughout the day and across the growing season. As a result, the property is composed of numerous microclimatic zones, each

presenting distinct conditions for grapevine development. Such variability is a defining characteristic of the Douro wine region and is one of the factors responsible for the production of wines with unique sensory attributes and terroir expression. The absence of a dominant aspect indicates that the terrain has been shaped by complex geological and erosional processes, resulting in slopes facing multiple directions. This creates a mosaic of environmental conditions throughout the estate. South-facing slopes generally receive the highest levels of solar radiation, particularly during the summer months. These areas tend to experience warmer temperatures, faster grape maturation, and increased sugar accumulation. Consequently, grapes grown on south-facing slopes often reach harvest maturity earlier than those located on cooler exposures. North-facing slopes exhibit contrasting environmental conditions. Due to reduced solar exposure, these areas typically maintain lower temperatures and higher soil moisture levels. Grapes cultivated on north-facing slopes generally ripen more slowly, retaining higher acidity and developing more complex aromatic profiles. These characteristics are particularly valuable in regions experiencing increasing temperatures associated with climate change, as cooler sites help preserve grape quality and balance. East-facing slopes receive sunlight during the morning hours when temperatures are relatively mild. This exposure promotes gradual warming of the vine canopy and reduces the duration of moisture retention on leaves, thereby decreasing the risk of fungal diseases. West-facing slopes, in contrast, are exposed to stronger afternoon sunlight and higher temperatures. While this can enhance ripening, excessive afternoon heat may increase water stress and affect berry composition during particularly hot growing seasons. The heterogeneous aspect distribution observed at Quinta da Senhora da Graça creates a range of thermal environments across the property. These differences influence vine vigor, canopy development, flowering, fruit set, veraison, and harvest timing. The resulting spatial variability requires careful management to ensure that each vineyard block receives appropriate cultural practices according to its specific environmental conditions.

Aspect also affects evapotranspiration rates and water availability within the vineyard. Slopes receiving greater solar radiation generally experience higher evaporation rates, leading to reduced soil moisture and increased vine water demand. Conversely, cooler and more shaded slopes may conserve moisture for longer periods. Understanding these patterns is essential for optimizing irrigation strategies, particularly in Mediterranean climates where summer drought conditions are common. In addition to influencing grapevine physiology, aspect plays an important role in pest and disease dynamics. Temperature and humidity conditions associated with different slope orientations affect the life cycles of vineyard pests and pathogens. For example, warmer exposures may accelerate the development of certain insect pests, while cooler and more humid areas may favor fungal diseases. According to Aranha (2026), slope orientation influences air circulation, sunlight interception, and microclimatic conditions that directly affect pest behavior and population distribution within vineyards. The effectiveness of grape moth (*Lobesia botrana*) monitoring and mating disruption programs can also be influenced by aspect-related environmental conditions. Variations in temperature between different slope exposures may alter

insect emergence patterns, flight activity, and reproduction. Consequently, understanding the spatial distribution of slope orientations is important for designing efficient pest management strategies and optimizing the placement of monitoring traps and pheromone dispensers. From a precision viticulture perspective, aspect analysis provides valuable information for site-specific management. Geographic Information Systems (GIS) enable the creation of detailed aspect maps that help identify zones with similar environmental characteristics. These maps can be integrated with data on soil properties, elevation, vine vigor, and pest occurrence to support informed decision-making. Such approaches allow vineyard managers to optimize resource use, improve grape quality, and enhance environmental sustainability. The aspect model of Quinta da Senhora da Graça demonstrates that the vineyard is characterized by substantial spatial variability in slope orientation. Rather than being dominated by a single exposure, the property contains a complex mixture of north-, south-, east-, and west-facing slopes. This heterogeneity contributes to the formation of diverse microclimates that influence every stage of vineyard development, from vine growth and fruit maturation to pest management and harvesting operations. Ultimately, the analysis of aspect highlights the intricate relationship between terrain morphology and vineyard performance. The diverse slope orientations present within Quinta da Senhora da Graça create a dynamic agricultural environment that supports the production of high-quality grapes while presenting unique management challenges. Understanding these topographic influences is essential for the successful implementation of precision viticulture practices and the sustainable management of vineyards in the Douro region.

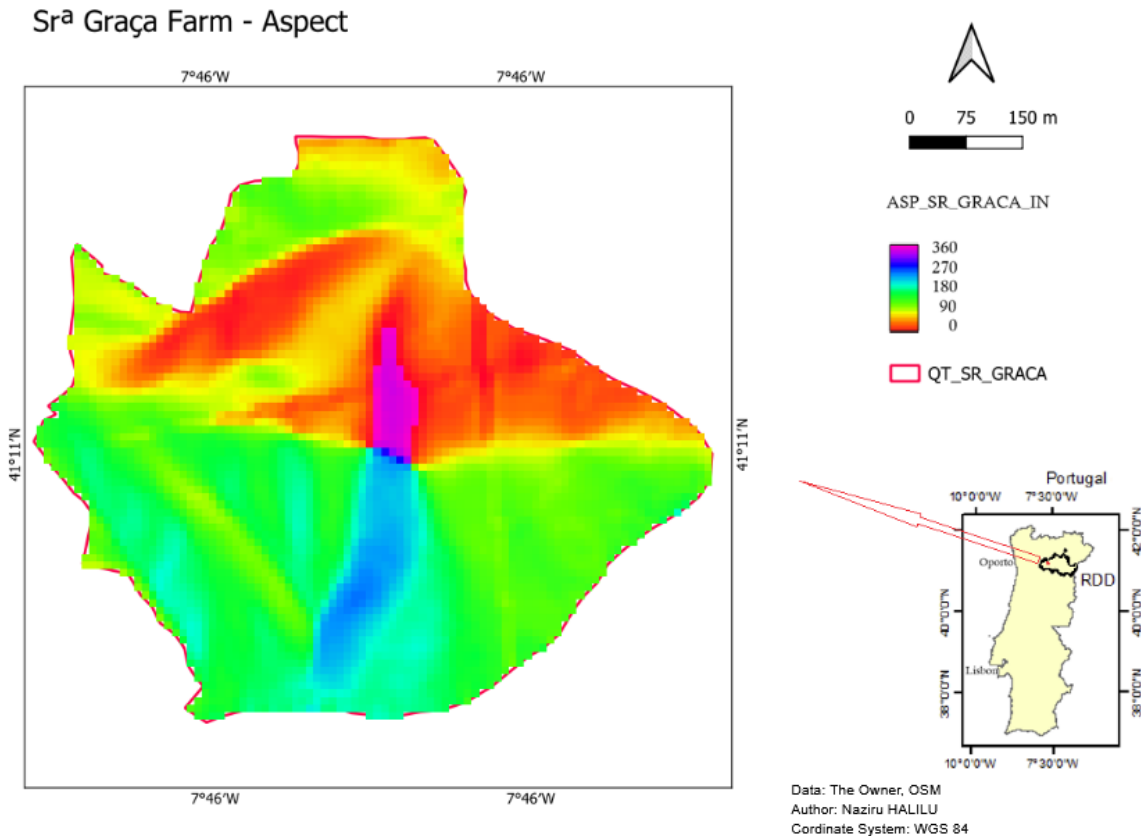


Figure 6. Aspect model of Quinta da Senhora da Graça property.

6.0 Land Use and Land Cover Distribution of Quinta da Senhora da Graça

Land use and land cover analysis is an essential component of precision agriculture and environmental management because it provides detailed information regarding the spatial organization of different activities and ecosystems within a property. At Quinta da Senhora da Graça, the analysis of land use demonstrates the strong agricultural character of the estate and highlights the dominant role of viticulture in shaping the landscape. The property occupies a total area of approximately 429,690.5 m² (42.97 ha) and exhibits a land-use pattern that is closely aligned with the historical and economic importance of wine production in the Douro Demarcated Region. The distribution of land use classes reveals that agricultural activities occupy the overwhelming majority of the estate. Agricultural land covers approximately 89.6% of the total property area, demonstrating the intensive utilization of the landscape for crop production. This predominance reflects the favorable environmental conditions of the region, including suitable climate, topography, and soil characteristics that support profitable viticultural production. The concentration of agricultural land also illustrates the long-standing relationship between human activities and the natural environment in the Douro Valley, where centuries of vineyard cultivation have transformed the landscape into one of the most recognizable wine-producing regions in the

world. Within the agricultural category, viticulture represents the principal land use. Vineyards occupy approximately 349,692.77 m², corresponding to about 90.8% of the total agricultural area and approximately 81.4% of the entire property. This dominance clearly demonstrates the economic specialization of Quinta da Senhora da Graça in grape production. The extensive vineyard coverage reflects the suitability of the estate for wine production and highlights the importance of viticulture as the primary source of income and agricultural activity within the property. The prevalence of vineyards is not only economically significant but also environmentally and culturally important. Viticulture has shaped the Douro landscape for centuries, creating terraces, retaining walls, and distinctive vineyard patterns that contribute to the region's recognition as a UNESCO World Heritage cultural landscape. The vineyards at Quinta da Senhora da Graça therefore represent both an agricultural resource and an integral component of the cultural heritage of the Douro Valley.

In addition to agricultural areas, forested land occupies approximately 9.7% of the total estate. Although relatively small compared to vineyard coverage, these forest areas perform important ecological functions. Forest vegetation contributes to biodiversity conservation, soil protection, carbon sequestration, and water regulation. Natural vegetation zones also provide habitat for beneficial insects, birds, and other wildlife species that can contribute to ecological balance within the vineyard ecosystem. The presence of forested areas is particularly important in integrated vineyard management systems because these areas can serve as ecological reservoirs that support natural enemies of agricultural pests. Beneficial insects originating from forest habitats may help suppress populations of vineyard pests, reducing dependence on chemical control measures. Furthermore, forested areas help stabilize slopes and reduce soil erosion, which is especially important in the steep terrain conditions characteristic of the Douro region. Human occupation areas account for only 0.7% of the total property area. These areas include residential buildings, storage facilities, machinery shelters, access roads, operational infrastructure, and other support structures necessary for agricultural activities. Although occupying a relatively small proportion of the estate, these facilities are essential for the efficient functioning of the vineyard operation. The limited extent of built-up areas reflects the predominantly agricultural nature of the property and demonstrates the emphasis placed on maintaining productive land for viticulture. The spatial organization of land use within the estate reflects both environmental constraints and management objectives. Vineyard plots are generally established in areas offering favorable conditions for grape cultivation, including suitable slopes, adequate solar exposure, and well-drained soils. Forested areas are often located in zones less suitable for cultivation due to steep gradients, shallow soils, or conservation requirements. This allocation of land uses helps maximize agricultural productivity while preserving important ecological functions. The analysis of land use patterns also provides valuable information for precision viticulture and sustainable land management. Geographic Information Systems (GIS) allow the detailed mapping and monitoring of different land-use classes, enabling vineyard managers to evaluate spatial relationships between agricultural activities, natural vegetation, and infrastructure. Such information supports decision-making

related to resource allocation, pest management, erosion control, biodiversity conservation, and environmental sustainability. From a climatic perspective, the distribution of land-use classes influences local microclimatic conditions across the property. Forested areas contribute to humidity regulation, temperature moderation, and wind reduction, while vineyard areas influence heat absorption and evapotranspiration processes. These interactions create a dynamic landscape where land cover characteristics affect environmental conditions experienced by grapevines throughout the growing season.

The predominance of viticulture at Quinta da Senhora da Graça also reflects broader regional patterns observed throughout the Douro Valley. Wine production remains one of the most important economic activities in northern Portugal and serves as a major driver of rural development, employment, and cultural identity. The vineyard-dominated landscape contributes significantly to the international reputation of the region and supports a thriving wine tourism industry that attracts visitors from around the world. Furthermore, land-use analysis provides an important foundation for environmental monitoring and future planning. Changes in land cover over time can influence hydrological processes, biodiversity, carbon storage, and agricultural productivity. Continuous monitoring of land-use patterns therefore enables managers to assess the effectiveness of conservation measures and adapt management strategies to changing environmental and economic conditions. Overall, the land-use and land-cover distribution of Quinta da Senhora da Graça clearly demonstrates the agricultural specialization of the property, with viticulture representing the dominant land use. The combination of extensive vineyard coverage, strategically located forest areas, and limited human occupation zones reflects a landscape carefully adapted to the environmental conditions and economic objectives of the Douro wine region. This spatial organization supports both productive viticulture and the long-term sustainability of the estate while preserving the unique cultural and environmental values that characterize the Douro Valley.

Table 1. *Distribution of land-use and land-cover classes, occupied area, percentage coverage, and number of associated plots within Quinta da Senhora da Graça.*

Land Cover	Agriculture Parcels	Agriculture Area (m ²)	Forestry Parcels	Forestry Area (m ²)	Human Parcels	Human Area (m ²)	Total Parcels	Total Area (m ²)
Wine cellar	0	0	0	0	1	975.8	1	975.8
General agriculture	4	6425.0	0	0	0	0	4	6425.0
Support for the vineyard	0	0	0	0	1	682.1	1	682.1
Grove	0	0	6	4505.3	0	0	6	4505.3
Riparian forest	0	0	2	35121.4	0	0	2	35121.4
Housing	0	0	0	0	2	1506.2	2	1506.2
Olive grove	5	27845.3	0	0	0	0	5	27845.3
Bare soil	1	3250.3	0	0	0	0	1	3250.3
Vineyard	29	349477.5	0	0	0	0	29	349477.5
Grand Total	39	386998.141	8	39626.653	4	3164.116	51	429690.5

Sr^a Graça Farm - Land cover

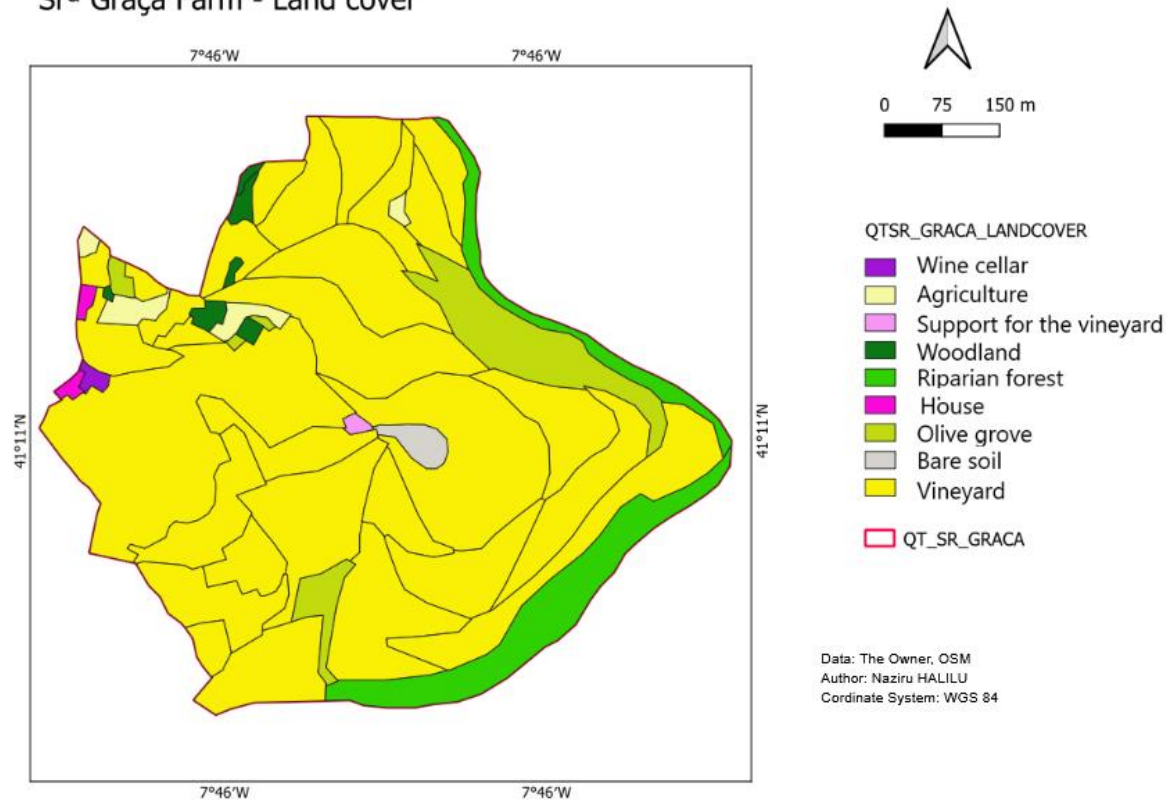


Figure 6- Land Use and Occupation Chart of the Qt.^a da Sr.^a da Graça.

7.0 Vineyard Varietal Composition

The estate exhibits significant varietal diversity, with approximately ten grapevine varieties cultivated throughout the vineyard. The dominant varieties include Touriga Nacional, Touriga Franca, and Tinta Roriz, which are among the most important grape cultivars used in Douro wine production. Touriga Nacional occupies approximately 42,932.9 m², while Touriga Franca occupies 44,883.1 m². Tinta Roriz also occupies a significant area of 37,225.3 m². According to Aranha (2026), these varieties are highly adapted to the climatic and topographic conditions of the Douro region. Red grape varieties dominate the viticultural landscape of the property, occupying approximately 187,543.9 m², corresponding to 53.7% of the vineyard area. White grape varieties such as Côdega de Larinho, Malvasia Fina, Rabigato, and Viosinho occupy approximately 75,602.8 m², representing 21.6% of the vineyard surface. Mixed-variety vineyard plots occupy approximately 86,330.8 m², corresponding to 24.7% of the total vineyard area. An important characteristic of the property is the presence of mixed-variety vineyard plots, which account for approximately one-quarter of the vineyard surface. Mixed vineyards are common in traditional Douro viticulture and contribute to biodiversity, resilience, and wine complexity. According to [Aranha \(2026\)](#), varietal diversity may influence pest susceptibility, grape maturation timing, and

disease development. Different grape varieties may also respond differently to climatic stress and pest pressure. Consequently, varietal mapping through GIS provides valuable information for precision viticulture management and targeted phytosanitary interventions. This farm has approximately 11 vine varieties installed, with a predominance of mixed grape varieties (Mistura de castas), which occupy approximately 86,330.8 m² (25.0%) of the vineyard area, as shown in Figure 2 and described in Table 2. The vineyard presents a combination of white, red, and mixed-variety grape cultivars, reflecting the traditional viticultural diversity characteristic of the Douro Demarcated Region. Among the red grape varieties, Touriga Nacional and Touriga Franca occupy significant areas, while Viosinho, Malvasia Fina, and Côdega de Larinho are among the main white grape varieties cultivated on the property. In terms of land use and land cover (Fig. 7), the property is predominantly agricultural, with agriculture occupying approximately 386,998.1 m², corresponding to 89.6% of the total area. Forest areas occupy approximately 39,626.7 m² (9.7%), while human occupation areas, including housing, winery facilities, and vineyard support infrastructures, represent approximately 3,164.1 m² (0.7%) of the total property area. Agricultural activity is mainly dedicated to viticulture, with vineyards occupying approximately 349,477.5 m², corresponding to nearly 81.3% of the total farm area and approximately 90.3% of the agricultural land. Other agricultural land uses include olive groves (27,845.3 m²), general agriculture (6,425.0 m²), and bare soil areas (3,250.3 m²).

Sr^a Graça Farm - Vine types

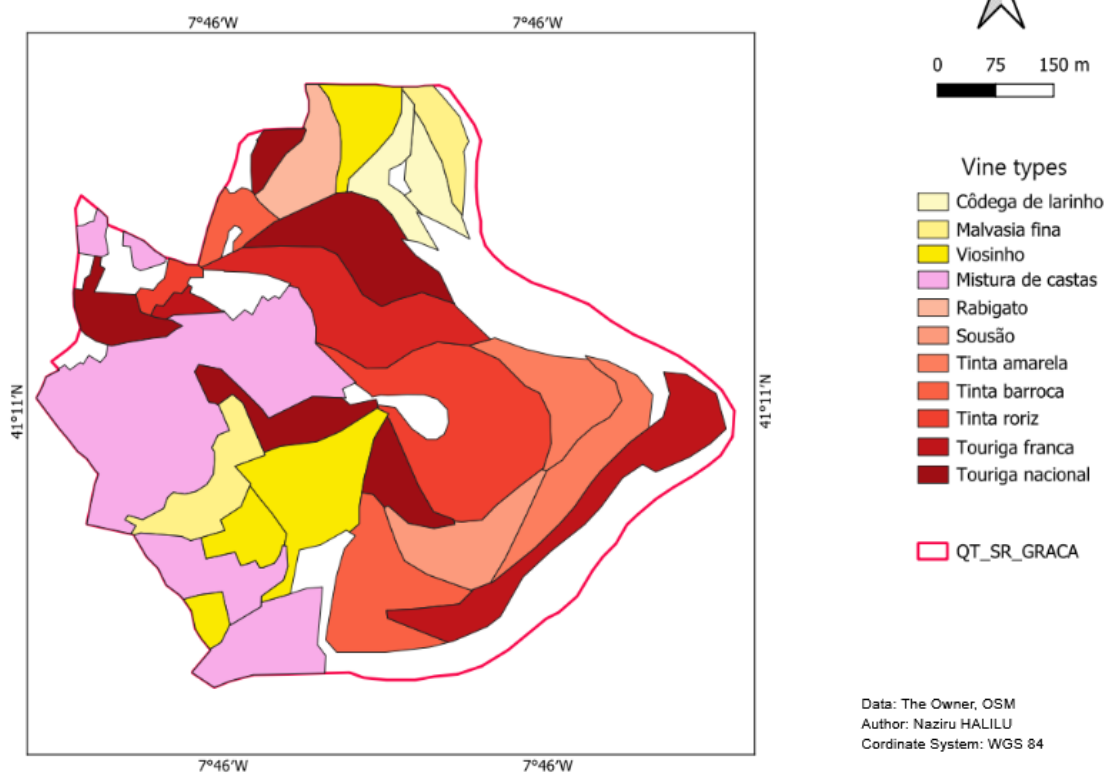


Figure 7- Caste chart of the Qt.^a da Sr.^a da Graça.

The forested areas are mainly composed of riparian forest vegetation (35,121.4 m²) and grove vegetation (4,505.3 m²), which contribute to ecological diversity within the property. Human occupation includes a wine cellar (975.8 m²), housing facilities (**1,506.2 m²**), and vineyard support infrastructure (682.1 m²). The detailed distribution of land use and land cover classes is presented in Table 1 for further analysis.

Table 2. Vineyard Varieties and Pheromone Diffuser Estimation

Vine Type	Total (m ²)	Area (%)	Diffusers Min	Diffusers Max	Labor Hours Min	Labor Hours Max	Cost (€) Min	Cost (€) Max
Códaga de larinho	12632.4	3.8	442	505	3.7	4.2	96.5	110.3
Malvasia Fina	16984.4	5.3	594	679	5.0	5.7	129.8	148.3
Mixed varieties	86330.8	25.0	3022	3453	25.2	28.8	659.7	754.0
Rabigato	9712.8	2.6	340	389	2.8	3.2	74.2	84.8
Sousão	12851.5	3.5	450	514	3.7	4.3	98.2	112.2
Tinta Amarela	25742.0	8.1	901	1030	7.5	8.6	196.7	224.8
Tinta Barroca	23909.1	6.8	837	956	7.0	8.0	182.7	208.8
Tinta Roriz	37225.3	10.1	1303	1489	10.9	12.4	284.5	325.1
Touriga Franca	44883.1	13.4	1571	1795	13.1	15.0	343.0	392.0
Touriga Nacional	42932.9	13.5	1503	1717	12.5	14.3	328.1	374.9
Viosinho	36273.2	8.0	1270	1451	10.6	12.1	277.2	316.8
Total	349477.5	100.0	12232	13979	101.9	116.5	2670.6	3052.1

Table 3. Vineyard varietal composition and spatial distribution at Quinta da Senhora da Graça

Vine Type	White Parcels	White (m ²)	Area	Red Parcels	Red (m ²)	Area	Mixed Parcels	Red	Mixed (m ²)	Red	Area	Total Parcels
Côdega de Larinho	2	12,632.4		0	0.0		0		0.0			2
Malvasia Fina	2	16,984.4		0	0.0		0		0.0			2
Mistura de Castas	0	0.0		0	0.0		5		86,330.8			5
Rabigato	1	9,712.8		0	0.0		0		0.0			1
Sousão	0	0.0		1	12,851.5		0		0.0			1
Tinta Amarela	0	0.0		2	25,742.0		0		0.0			2
Tinta Barroca	0	0.0		2	23,909.1		0		0.0			2
Tinta Roriz	0	0.0		2	37,225.3		0		0.0			2
Touriga Franca	0	0.0		3	44,883.1		0		0.0			3
Touriga Nacional	0	0.0		5	42,932.9		0		0.0			5
Viosinho	4	36,273.2		0	0.0		0		0.0			4
Total	9	75,602.8		15	187,543.9		5		86,330.8			29
Total (%)	31.0	21.6		51.7	53.7		17.2		24.7			100.0

8.0 Pest Monitoring and Mating Disruption Programme for *Lobesia botrana* in Quinta da Senhora da Graça Vineyard

8.1 Introduction to Vineyard Pest Management

Effective pest management is a fundamental component of sustainable viticulture, particularly in high-value wine-producing regions where grape quality directly influences economic returns. Among the most significant insect pests affecting vineyards in the Douro region is the European grapevine moth (*Lobesia botrana*), a species capable of causing severe damage to grape clusters through larval feeding activities. The pest attacks flowers and berries, creating wounds that facilitate secondary infections by fungal pathogens such as *Botrytis cinerea*, thereby reducing both yield and fruit quality.

To minimize the environmental impacts associated with conventional pesticide applications and to comply with modern integrated pest management (IPM) principles, the Quinta da Senhora da Graça vineyard adopted a mating disruption strategy based on sexual confusion. This technique uses synthetic pheromone diffusers to interfere with the ability of male moths to locate females, thereby reducing mating success and suppressing pest populations. In addition, a monitoring programme based on delta traps was implemented to assess pest population dynamics and support decision-making regarding pest management interventions. The vineyard occupies a total area of 349,477.5 m² (34.95 ha), consisting of several grape varieties distributed across white, red, and mixed vineyard blocks. The implementation of the mating disruption programme and the associated monitoring activities were designed according to the spatial distribution of these vineyard parcels.

8.2 Implementation of the Sexual Confusion Method

The sexual confusion method was implemented using BIOOtwi[®]n L pheromone diffusers specifically developed for the management of *Lobesia botrana*. The methodology consisted of distributing pheromone dispensers throughout the vineyard at densities ranging between 350 and 400 diffusers per hectare. These densities are widely recognized as effective for disrupting the communication between male and female moths and reducing successful mating events. The number of diffusers required for each vineyard block was determined according to the area occupied by each grape variety. Larger vineyard blocks naturally required higher numbers of diffusers and greater labour inputs during installation. The mixed vineyard parcel known as Mistura de Castas occupied the largest area and consequently required the highest number of diffusers. Conversely, smaller vineyard parcels such as Rabigato required comparatively fewer diffusers and installation hours. The installation process was carried out manually at a rate of approximately 120 diffusers per hour. Labour requirements and costs were estimated using a labour rate of €7 per hour and a diffuser unit cost of €0.16. These parameters enabled the estimation of minimum and maximum implementation costs corresponding to diffuser densities of 350 and 400 diffusers per hectare, respectively.

8.3 Estimation of Diffuser Requirements, Labour Demand and Installation Costs

The calculation of diffuser requirements demonstrated a direct relationship between vineyard area and the resources required for mating disruption implementation. The total vineyard area required between 12,232 and 13,979 diffusers depending on the application density adopted. Correspondingly, labour requirements ranged from 101.9 to 116.5 working hours. The total implementation cost varied between €2,670.6 and €3,052.1. These costs included both diffuser acquisition and installation labour. The results indicate that mating disruption can be implemented across the entire vineyard with a predictable cost structure, allowing farm managers to plan annual pest management budgets effectively.

Table 4: Estimated Costs Associated with the Implementation of Sexual Confusion in Quinta da Senhora da Graça Vineyard

Vine Type	Diffusers (Min)	Diffusers (Max)	Man Min	Labour (h)	Man Max	Labour (h)	Cost Min	Cost Max
Côdega de Larinho	442	505	3.7		4.2		96.5	110.3
Malvasia Fina	594	679	5.0		5.7		129.8	148.3
Mistura de Castas	3022	3453	25.2		28.8		659.7	754.0
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Tinta Roriz	1303	1489	10.9		12.4		284.5	325.1
Touriga Franca	1571	1795	13.1		15.0		343.0	392.0
Touriga Nacional	1503	1717	12.5		14.3		328.1	374.9
Viosinho	1270	1451	10.6		12.1		277.2	316.8
Total	12,232	13,979	101.9		116.5		2,670.6	3,052.1

8.4 Spatial Monitoring of *Lobesia botrana* Using Delta Traps

Although mating disruption significantly reduces moth populations, continuous monitoring remains essential for evaluating programme effectiveness and detecting potential pest outbreaks. For this purpose, twenty-five delta traps were installed throughout the vineyard. The traps were georeferenced and strategically distributed to ensure homogeneous spatial coverage of the entire property. The distribution of the traps enabled representative monitoring of all vineyard sectors and grape varieties. By maintaining an even spatial arrangement, the monitoring system increased the probability of detecting pest activity throughout the production area. The georeferenced trap locations also facilitated the integration of monitoring data within Geographic Information Systems (GIS), allowing the visualization of pest dynamics and spatial variability across the vineyard. The monitoring programme focused on identifying the principal flight periods of *Lobesia botrana*. Trap inspections were conducted during April, June, and August, corresponding to critical stages in the pest life cycle and reflecting the typical flight curves observed within the region. Information obtained from these monitoring activities supports pest risk estimation and contributes to the optimization of future management interventions.

8.5 Estimation of Delta Trap Installation and Monitoring Costs

The economic assessment of the monitoring programme included trap acquisition, installation, and periodic monitoring activities. A total of twenty-five delta traps were purchased at a unit cost of €3.20 per trap. Installation operations were performed at a rate of four traps per hour, while monitoring activities were conducted at a rate of two traps per hour. Labour costs were estimated at €7 per hour. The results show that trap acquisition accounted for €80.00 of the total expenditure. Installation required 6.25 labour hours, resulting in an additional cost of €43.75. Each monitoring campaign required 12.5 labour hours, corresponding to €87.50 per monitoring event. Considering three monitoring campaigns during the growing season, the total monitoring cost reached €262.50. Consequently, the complete trap-based monitoring programme generated a total expenditure of €386.25. Compared with the total cost of the mating disruption programme, this monitoring component represents a relatively small investment while providing valuable information for pest management decision-making.

Table 5: Costs of Installation and Monitoring of Delta Traps

Item	Calculation	Cost (€)
Trap Cost	$25 \text{ traps} \times \text{€}3.20$	80.00
Trap Installation	$(25 \text{ traps} \div 4 \text{ traps h}^{-1}) \times \text{€}7 \text{ h}^{-1}$	43.75
Trap Survey (per monitoring action)	$(25 \text{ traps} \div 2 \text{ traps h}^{-1}) \times \text{€}7 \text{ h}^{-1}$	87.50
Trap Survey (3 monitoring actions)	$\text{€}87.50 \times 3$	262.50
Total Installation + Monitoring	$80.00 + 43.75 + 262.50$	386.25

8.6 Monitoring Programme

The monitoring programme provided a structured framework for assessing *Lobesia botrana* activity throughout the vineyard growing season. The combination of mating disruption and systematic monitoring represents an integrated and environmentally sustainable approach to vineyard pest management. By reducing dependence on chemical insecticides and emphasizing preventive control measures, the programme contributes to both economic sustainability and environmental protection. The monitoring network ensured adequate spatial representation of the vineyard, while the repeated observations during the critical months of April, June, and August allowed the characterization of pest population dynamics throughout the production cycle. Furthermore, the cost analysis demonstrated that the monitoring programme can be implemented with a relatively modest financial investment compared with the overall value of vineyard production.

Table 6: Monitoring Programme

Parameter	Value
Number of Delta Traps	25
Number of Monitoring Periods	3 (April, June and August)
Cost per Trap	€3.20
Installation Rate	4 traps h ⁻¹
Monitoring Rate	2 traps h ⁻¹
Labour Cost	€7 h ⁻¹
Total Monitoring Programme Cost	€386.25

The implementation of the mating disruption strategy using BIOOtwi[®] L pheromone diffusers constituted a comprehensive preventive management approach for controlling *Lobesia botrana* within the Quinta da Senhora da Graça vineyard. Based on the vineyard area of 34.95 ha, diffuser

requirements ranged from 12,232 to 13,979 units, with total implementation costs varying between €2,670.6 and €3,052.1. The largest investment was associated with vineyard blocks occupying the greatest area, particularly the Mistura de Castas parcel. The complementary monitoring programme, based on twenty-five georeferenced delta traps, enabled continuous assessment of pest population dynamics during the three principal monitoring periods. The total cost of the monitoring system was estimated at €386.25, representing a cost-effective mechanism for supporting integrated pest management decisions. Overall, the combined use of mating disruption and systematic monitoring provides an environmentally responsible, economically viable, and technically effective strategy for sustainable vineyard management.

9.0 Setting of Delta Traps for Monitoring of *Lobesia botrana*

The installation of delta traps constituted a fundamental component of the integrated pest management strategy adopted in the Quinta da Senhora da Graça vineyard. While the mating disruption programme aimed to reduce the reproductive success of *Lobesia botrana* populations through pheromone-based sexual confusion, the delta trap network provided a reliable monitoring system for evaluating pest activity throughout the production season. The primary objective of trap installation was to monitor fluctuations in moth populations, identify peak flight periods, assess the effectiveness of the mating disruption programme, and support informed management decisions.

The setting of traps was carried out following the principles of representative spatial sampling and uniform coverage of the vineyard area. A total of twenty-five delta traps were strategically distributed across the vineyard property, which covers approximately 34.95 hectares. The selected number of traps ensured that all vineyard sectors were adequately represented, including blocks planted with white grape varieties, red grape varieties, and mixed vineyard parcels. The distribution pattern was designed to minimize monitoring gaps and maximize the probability of detecting moth activity regardless of its location within the vineyard. Prior to installation, each trap location was carefully selected using geographic information system (GIS) tools and field reconnaissance. The selected points were georeferenced to ensure accurate positioning and facilitate future monitoring activities. Georeferencing also allowed the trap network to be integrated into the vineyard's spatial database, enabling the visualization of pest distribution patterns and supporting the development of spatially explicit management strategies. The trap locations were recorded using Global Navigation Satellite System (GNSS) devices to ensure consistency in monitoring throughout the growing season. The delta traps were installed at suitable positions within the vineyard canopy where moth activity was expected to be highest. Traps were suspended at approximately bunch height to maximize exposure to adult moths while minimizing interference from vineyard operations. Care was taken to avoid placing traps too close to vineyard borders, roads, or other structures that could influence moth movement patterns and compromise the representativeness of the monitoring results. Adequate spacing was maintained between traps to reduce overlap in attraction zones and ensure independent sampling of pest populations.

Each trap was equipped with pheromone lures specifically designed for attracting male *Lobesia botrana* moths. The pheromone attractants mimic the natural sex pheromones released by female moths, thereby enabling the capture of male individuals and providing an indirect measure of pest population density. The use of pheromone-based monitoring is widely recognized as an efficient and environmentally friendly technique for vineyard pest surveillance because it provides early warning information without affecting non-target organisms. The monitoring programme was structured around three key observation periods corresponding to the principal flight generations of *Lobesia botrana* in the region. Trap inspections were conducted during April, June, and August, which represent critical periods in the pest life cycle. During each monitoring event, captured moths were counted and recorded to assess population dynamics and estimate pest pressure. These observations allowed the identification of temporal trends in moth abundance and provided valuable information regarding the effectiveness of the mating disruption strategy implemented throughout the vineyard.

The homogeneous spatial distribution of the twenty-five traps ensured comprehensive monitoring coverage across the entire vineyard area. This arrangement enabled the detection of localized population increases and facilitated comparisons among different vineyard blocks and grape varieties. Furthermore, the georeferenced trap data provided opportunities for subsequent spatial analyses, including the generation of distribution maps, interpolation surfaces, and hotspot identification using GIS techniques. Such analyses contribute to a better understanding of pest ecology and support precision agriculture approaches for vineyard management. The installation process itself required careful planning and labour allocation. Based on an installation rate of four traps per hour, the deployment of the twenty-five traps required approximately 6.25 labour hours. Considering a labour cost of €7 per hour, the total installation cost amounted to €43.75. In addition to installation expenses, the acquisition cost of the traps was estimated at €80.00, assuming a unit price of €3.20 per trap. These costs represent a relatively small investment when compared to the economic benefits associated with timely pest detection and effective management.

Once installed, the traps formed the basis of a continuous surveillance system throughout the growing season. Regular inspections ensured that pheromone lures remained effective and that captured insects were accurately counted and recorded. Monitoring activities were conducted at a rate of approximately two traps per hour, requiring 12.5 labour hours per survey event. The information obtained through this process supported risk assessment procedures and enabled vineyard managers to evaluate whether additional interventions were necessary. Overall, the setting of delta traps represented a critical element of the vineyard's integrated pest management programme. The combination of strategic trap placement, georeferenced monitoring, and systematic data collection provided a robust framework for assessing *Lobesia botrana* populations throughout the season. By enabling early detection of pest activity and supporting evidence-based decision-making, the trap network contributed significantly to the sustainability, efficiency, and economic viability of vineyard production at Quinta da Senhora da Graça.

Sr^a Graça Farm - Traps' location

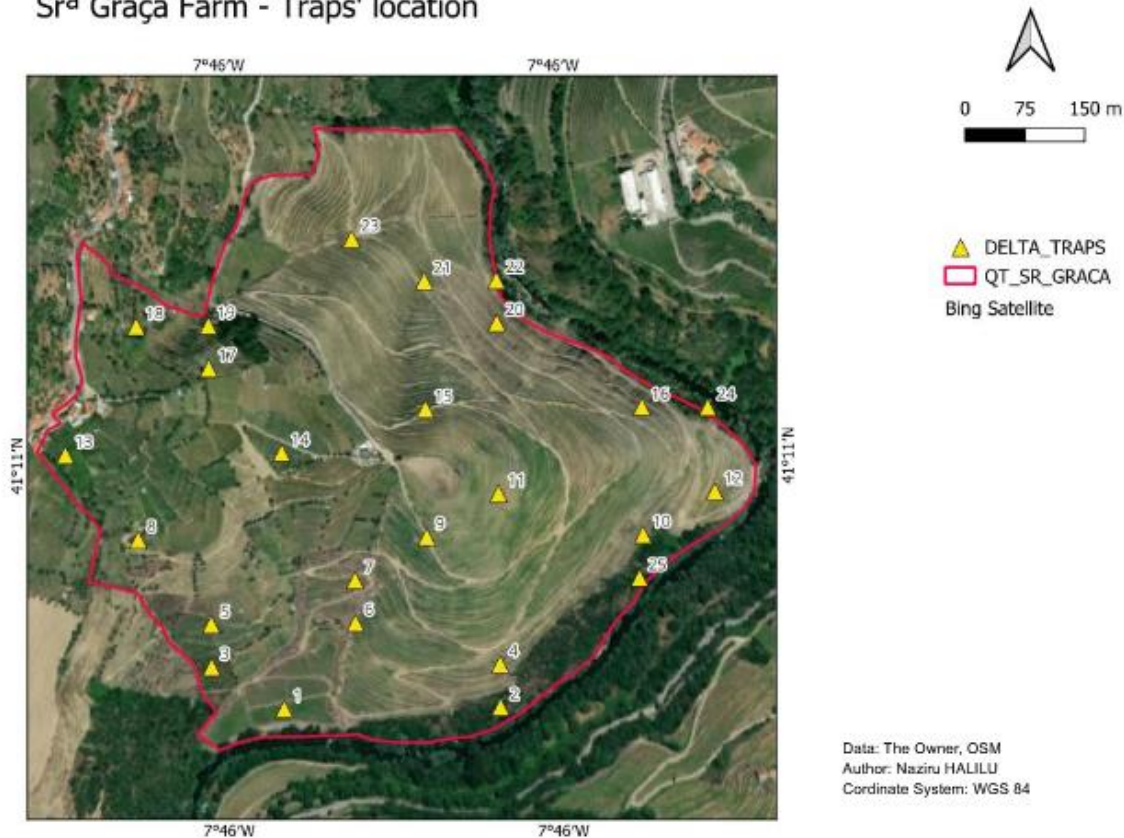


Figure 8- Georeferenced location of the delta traps installed in the Qt.^a da Sr.^a da Graça

10.0 Spatial Analysis of *Lobesia botrana* Captures and Vineyard Damage Assessment

10.1 Monitoring of *Lobesia botrana* Throughout the Vegetative Cycle

The monitoring of *Lobesia botrana* populations and the associated vineyard damage was conducted throughout the vegetative cycle of the grapevine in order to evaluate the effectiveness of the mating disruption programme implemented at Quinta da Senhora da Graça. Monitoring activities were performed during three critical periods corresponding to the major generations of the pest. These periods included April–May, June–July, and August–September, covering the principal flight periods and subsequent larval infestation stages. The monitoring strategy combined pheromone trap captures with direct field observations of damage symptoms. This integrated approach enabled the establishment of relationships between adult moth populations and the subsequent occurrence of larval damage within different vineyard sectors. By evaluating both captures and observed damage, it was possible to assess the spatial effectiveness of the sexual confusion programme and identify areas where pest pressure remained significant despite

management interventions. To facilitate accurate data collection and spatial analysis, all occurrences of vineyard damage were georeferenced in the field using a Geographic Information Systems (GIS) project developed through the QField mobile application. The georeferenced observations were subsequently integrated into the GIS database, allowing the visualization of spatial patterns and the comparison of pest activity among different periods of the production season.

10.2 Spatial Distribution of Captures and Damage During the First Flight Period (April–May)

The spatial relationship between *Lobesia botrana* captures recorded during April and the damage observed during May is presented in Figure 9. This period corresponds to the first significant flight generation of the pest and represents the initial stage of population development within the vineyard. The results demonstrate that captures recorded in the delta traps were generally low to moderate, reflecting the early-season emergence of adult moths. Nevertheless, the spatial distribution of the observed damage revealed a close relationship with the locations where captures occurred. Damage symptoms, including berry perforation and feeding injuries, were predominantly concentrated near traps that registered moth activity, indicating that adult captures served as a reliable indicator of subsequent larval infestation. The analysis also revealed localized clusters of damage within specific sectors of the vineyard. These concentrations suggest that certain areas may provide more favourable environmental conditions for pest development, including shelter, microclimatic advantages, or proximity to alternative host plants. Consequently, the April–May monitoring period provided valuable information regarding the initial establishment of *Lobesia botrana* populations within the property and highlighted areas requiring closer surveillance during subsequent generations.

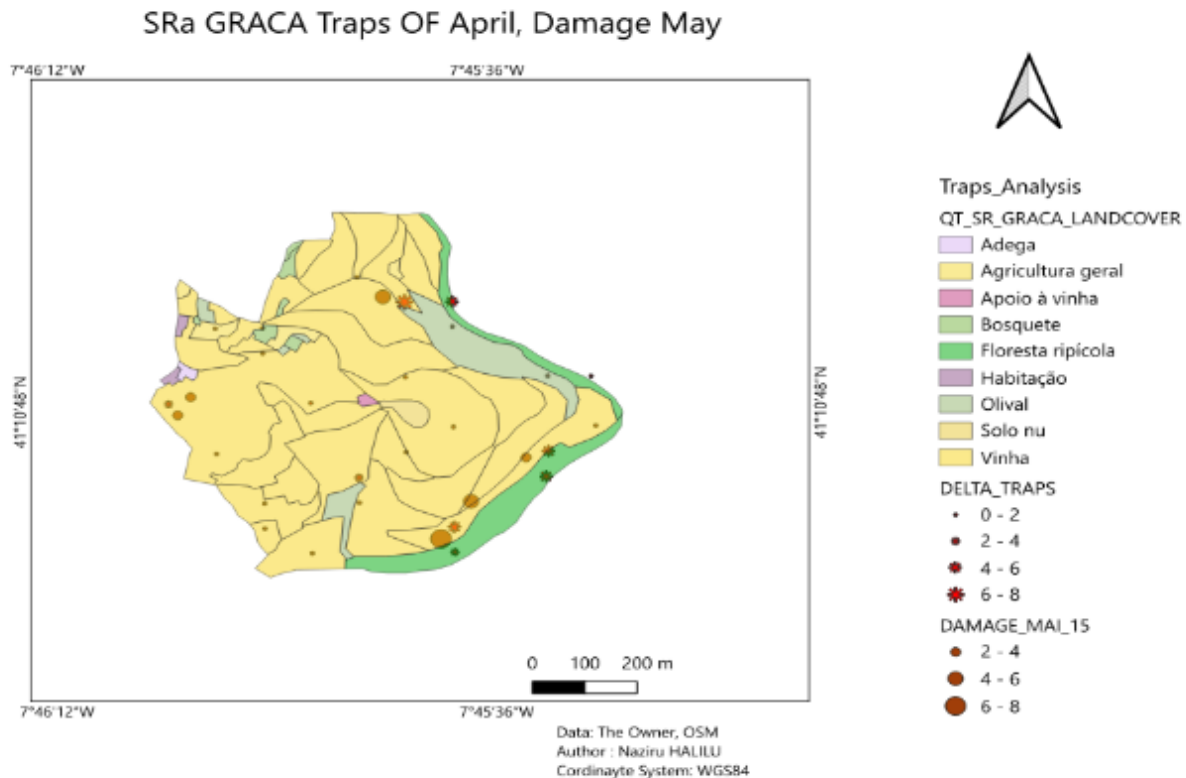


Figure 9. Spatial distribution of *Lobesia botrana* captures in April and the damage observed in May at Quinta da Senhora da Graça.

10.3 Spatial Distribution of Captures and Damage During the Second Flight Period (June–July)

The spatial relationship between captures recorded in June and the damage observed in July is illustrated in Figure 10. During this period, the vineyard experienced the second flight generation of *Lobesia botrana*, which is typically associated with increased population density and a higher risk of crop damage. The joint analysis of Figures 9 and 10 indicates a noticeable increase in the concentration of captures during the second flight period. Several vineyard sectors exhibited higher trap catches compared with the first generation, reflecting the natural population growth of the pest during favourable climatic conditions. Correspondingly, the distribution of damage observed in July became more extensive and more concentrated around areas where higher capture levels were recorded. This increase in pest activity demonstrates the importance of continuous monitoring throughout the growing season. The second generation is often considered one of the most economically important phases of *Lobesia botrana* development because larval feeding occurs directly on developing grape clusters. Consequently, the observed increase in damage during July

suggests that some local populations remained active despite the presence of pheromone diffusers. The spatial concentration of both captures and damage during this period further emphasizes the value of GIS-based monitoring. Through spatial analysis, vineyard managers can identify specific management zones where additional preventive measures may be required to reduce pest pressure and minimize production losses.

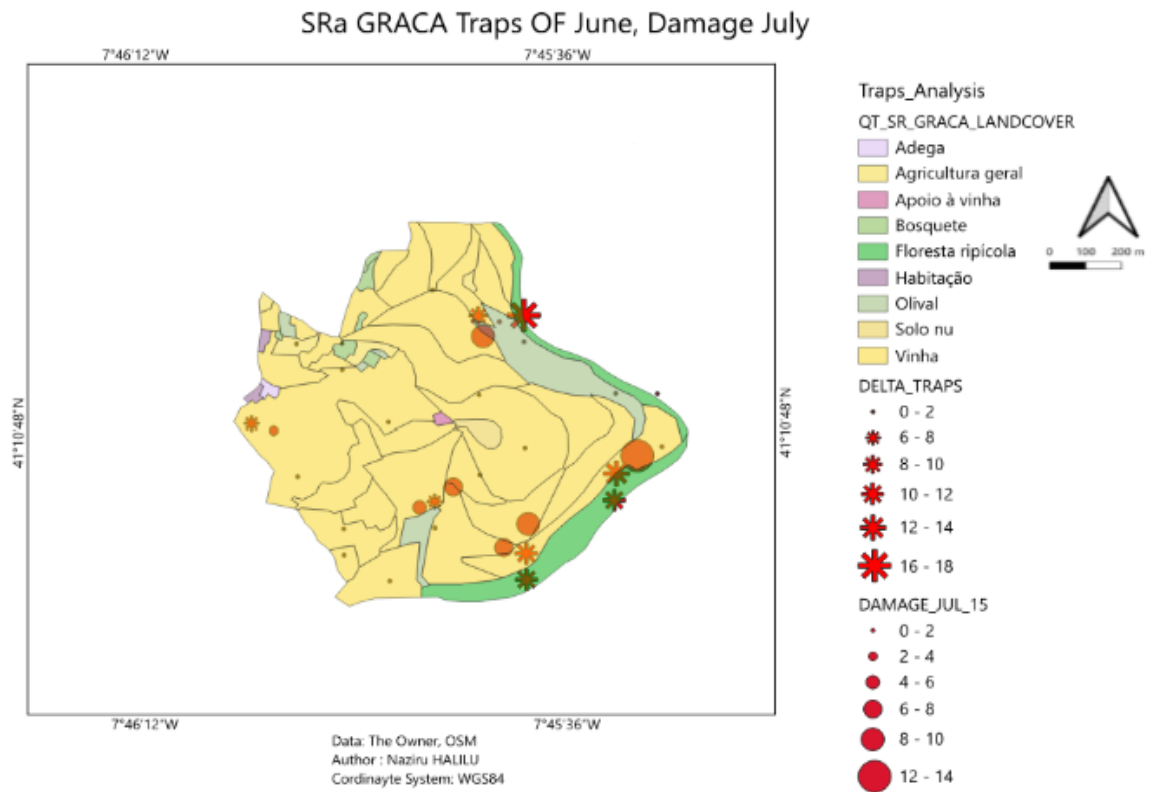


Figure 10. Spatial distribution of *Lobesia botrana* captures in June and the damage observed in July at Quinta da Senhora da Graça.

10.4 Spatial Distribution of Captures and Damage During the Third Flight Period (August–September)

The third monitoring period included captures recorded during August and damage observations conducted in September. The corresponding spatial distribution is presented in Figure 11. This phase represents the final major generation of *Lobesia botrana* during the production season and is frequently associated with significant economic losses due to berry maturation and increased susceptibility to fungal infections. The analysis revealed a stronger expression of pest activity during the third flight generation. Although the mating disruption programme continued to suppress large-scale population outbreaks, certain vineyard sectors exhibited persistent capture activity and localized concentrations of damage. The relationship between trap captures and

observed injury remained evident, confirming the effectiveness of pheromone traps as indicators of pest pressure. The damage observed during September was characterized not only by direct feeding injuries but also by the increased occurrence of berry necrosis and secondary rot infections. These symptoms result from larval feeding wounds that facilitate pathogen entry into grape clusters. Consequently, damage during this stage has a particularly significant impact on grape quality and market value. The spatial persistence of damage in specific vineyard sectors suggests that local environmental conditions, vineyard structure, or external pest sources may influence the effectiveness of the mating disruption programme. These findings highlight the need for continuous evaluation of pest management strategies and the implementation of complementary control measures where necessary.

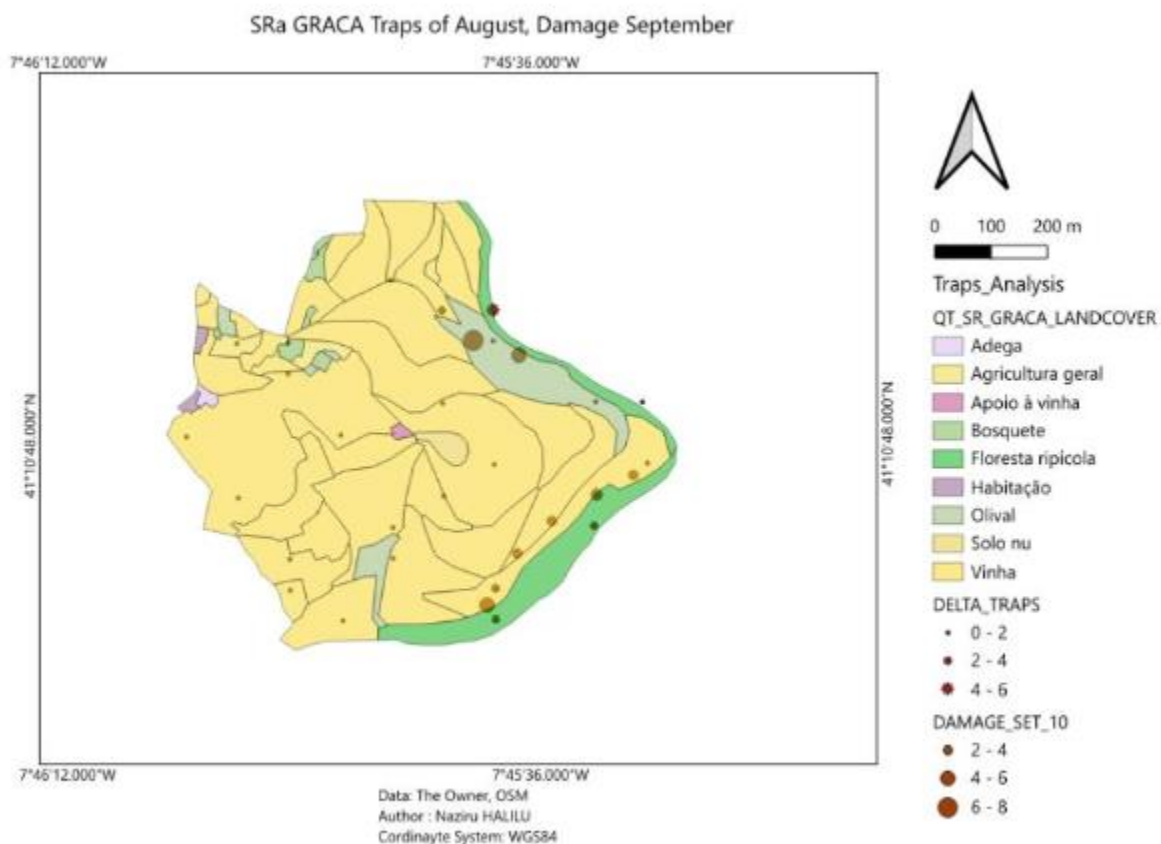


Figure 11. Spatial distribution of *Lobesia botrana* captures in August and the damage observed in September at Quinta da Senhora da Graça.

10.5 Spatial Variability in the Effectiveness of Sexual Confusion

The combined analysis of Figures 9, 10, and 11 demonstrates that the effectiveness of sexual confusion was not uniform throughout the vineyard. Although the overall population levels remained relatively controlled, some sectors consistently exhibited higher trap catches and greater damage intensity across multiple monitoring periods. The localized persistence of damage suggests that the effectiveness of mating disruption may vary spatially within the property. Both captures and damage tended to be concentrated preferentially in the same vineyard sectors throughout the monitoring programme. Several factors may contribute to this phenomenon, including higher local population pressure, vineyard edge effects, proximity to untreated neighbouring vineyards, surrounding vegetation that serves as pest reservoirs, and variations in vineyard topography and microclimatic conditions. Areas situated near external sources of infestation may experience continual immigration of adult moths, reducing the effectiveness of pheromone-based disruption. Similarly, variations in wind direction and terrain may influence pheromone dispersion patterns and affect the ability of the mating disruption system to maintain adequate coverage throughout the vineyard. The identification of these spatial patterns is essential for optimizing future management strategies. Rather than applying uniform interventions across the entire vineyard, site-specific management approaches can be developed to target areas with consistently higher pest pressure.

10.6 Relationship Between Trap Captures and Observed Damage

A clear relationship was observed between the spatial distribution of trap captures and the occurrence of vineyard damage. In most cases, damage was concentrated near traps that recorded captures of *Lobesia botrana*. This pattern demonstrates that pheromone trap data provide valuable predictive information regarding future infestation risk. The ability to establish a spatial connection between adult moth activity and larval damage represents one of the principal advantages of integrating GIS technology with pest monitoring programmes. Through the analysis of georeferenced data, vineyard managers can identify emerging hotspots of infestation before severe damage becomes widespread. This information facilitates timely interventions and contributes to more efficient resource allocation. The results obtained in this study demonstrate that delta traps not only provide information regarding pest population dynamics but also serve as effective tools for identifying areas requiring additional protective measures. Consequently, trap-based monitoring should be considered an essential component of integrated pest management programmes for *Lobesia botrana*.

10.7 Recommendations for Biological and Chemical Control Measures

Based on the observed relationship between captures and damage, the implementation of complementary control measures is recommended in areas exhibiting persistent pest activity. Where captures indicate the presence of newly emerging larvae and damage levels remain relatively low, the application of *Bacillus thuringiensis* (Bt) is recommended. This biological

insecticide acts as a larval control agent and is particularly effective when applied during the early larval instars. Commercial products containing *Bacillus thuringiensis*, such as DOCTRIN 32, KIPLANT BT-KURE, and VIJAY 32, may be applied immediately following the detection of the first caterpillars. Effective application requires thorough wetting of the vegetation to ensure adequate coverage of grape clusters and other plant organs susceptible to infestation. Higher application rates may be justified under conditions of elevated pest pressure. Where infestation progresses and visible symptoms of necrosis, berry damage, or rot become apparent, stronger interventions may be necessary. In such situations, the application of registered insecticides containing cypermethrin may be considered according to local regulations and integrated pest management guidelines. Products such as CYTHRIN MAX, CYPRESS 100 EC, CYPRESS, CYTHRIN 10 EC, and CYTHRIN OLIVO may be used when pest populations reach economic thresholds and significant crop losses are anticipated. The selection of appropriate control measures should always be based on monitoring results, field observations, and integrated pest management principles to ensure both economic efficiency and environmental sustainability. The results obtained demonstrate that the integration of delta trap monitoring, GIS-based spatial analysis, and field damage assessment constitutes a powerful tool for evaluating the effectiveness of mating disruption programmes against *Lobesia botrana*. The georeferenced monitoring system enabled the identification of spatial patterns in pest activity and revealed a strong relationship between trap captures and subsequent vineyard damage. The analysis further showed that although sexual confusion effectively reduced overall pest pressure, localized areas of persistent infestation remained present throughout the season. These findings emphasize the importance of combining preventive techniques with continuous monitoring and targeted interventions. The adoption of site-specific management strategies based on spatial analysis can improve pest control efficiency, reduce unnecessary pesticide applications, and contribute to the long-term sustainability of vineyard production systems.

11.0 Spatial Relationship Between Captures and Damage

11.1 Spatial Relationship Between *Lobesia botrana* Captures and Vineyard Damage

The spatial analysis conducted throughout the monitoring programme revealed a strong relationship between the captures of *Lobesia botrana* in pheromone delta traps and the occurrence of damage within the vineyard. The integration of georeferenced trap data with field observations of damage allowed a detailed assessment of how pest populations were distributed across the property and how these populations translated into economic injury on grape clusters. The results consistently demonstrated that damage tended to occur in proximity to traps where captures had been recorded. Areas with higher trap catches generally exhibited a greater incidence of larval feeding injuries, berry perforation, and subsequent secondary infections. This relationship confirms that adult moth captures can serve as reliable indicators of future infestation risk. Consequently, pheromone traps provide an effective early warning system that enables vineyard managers to anticipate areas where intervention may be required before severe damage develops.

The temporal sequence observed during the monitoring programme further supports this relationship. Captures recorded during each flight period were followed by damage observations in subsequent weeks, reflecting the biological development of the pest from adult moths to damaging larval stages. The consistency of this pattern throughout the monitoring season highlights the value of combining entomological monitoring with spatial analysis techniques. Such an approach allows managers not only to identify pest presence but also to predict where economic damage is most likely to occur. Furthermore, the spatial analysis identified specific vineyard sectors where both captures and damage repeatedly occurred throughout the season. These recurring hotspots indicate areas of persistent pest pressure and suggest the existence of favourable environmental conditions for pest establishment and reproduction. The identification of these zones provides opportunities for targeted management interventions and more efficient resource allocation.

11.2 Influence of Terrain Morphology on Pest Dispersion

Terrain morphology plays an important role in determining the spatial distribution and movement patterns of insect pests within agricultural landscapes. In vineyards, factors such as slope, elevation, aspect, and topographic position influence local microclimatic conditions, which in turn affect pest behaviour, survival, and reproduction. The study area is characterized by a heterogeneous terrain typical of vineyard systems established in hilly regions. Variations in elevation and slope create differences in temperature, solar radiation, wind exposure, and humidity. These environmental factors influence the activity and dispersion of *Lobesia botrana* adults, affecting their capacity to colonize different vineyard sectors. Lower landscape positions often accumulate higher humidity levels and may provide more favourable conditions for larval survival and disease development. Conversely, upper slopes may experience greater wind exposure, potentially influencing the dispersion of pheromone plumes released by mating disruption diffusers. Such variations can affect the effectiveness of the sexual confusion technique and contribute to the uneven distribution of pest populations across the vineyard. The spatial persistence of captures and damage observed in certain sectors may therefore be partially explained by topographic influences. Areas protected from strong winds may retain pheromone concentrations more effectively, whereas exposed locations may experience reduced disruption efficiency due to increased pheromone dilution. Similarly, sheltered areas may facilitate adult moth activity and increase local population densities. The incorporation of digital elevation models (DEMs) and terrain analysis within Geographic Information Systems provides valuable opportunities for understanding these relationships. By integrating pest monitoring data with topographic variables, vineyard managers can identify landscape features associated with increased pest risk and improve the spatial targeting of management practices.

11.3 Integrated Pest Management Strategies for *Lobesia botrana*

The results obtained from the monitoring programme demonstrate the importance of adopting an Integrated Pest Management (IPM) approach for the sustainable control of *Lobesia botrana*. IPM combines biological, cultural, mechanical, and chemical control methods to maintain pest populations below economically damaging thresholds while minimizing environmental impacts.

The mating disruption programme implemented through the use of BIOOtwinn® L pheromone diffusers constituted the primary preventive control measure. By interfering with the mating process of adult moths, the technique reduced the likelihood of successful reproduction and limited the growth of pest populations throughout the season. However, the persistence of localized infestations indicates that mating disruption alone may not provide complete protection under all conditions. Continuous monitoring using georeferenced delta traps therefore serves as an essential complementary component of the IPM programme. Monitoring enables the early detection of population increases and provides the information necessary for timely interventions. The combination of monitoring and spatial analysis allows vineyard managers to identify risk zones and prioritize management actions where they are most needed. Biological control measures should be considered as the first corrective intervention when captures indicate the presence of newly emerging larvae. Products based on *Bacillus thuringiensis* (Bt) provide an environmentally friendly alternative for controlling young caterpillars while preserving beneficial organisms. The use of Bt products is particularly effective when applied during the early larval stages and when guided by monitoring data. Where pest populations exceed economic thresholds and damage becomes severe, selective chemical control measures may be required. Such interventions should always be based on monitoring results and implemented according to integrated pest management principles to minimize environmental impacts and reduce the risk of pesticide resistance development. The combination of preventive, biological, and corrective measures provides a balanced and sustainable framework for managing *Lobesia botrana* populations while maintaining grape quality and vineyard profitability.

11.4 Role of Precision Agriculture Technologies in Vineyard Pest Management

Precision agriculture technologies are transforming modern vineyard management by enabling data-driven decision-making and site-specific interventions. In the present study, Geographic Information Systems (GIS), Global Navigation Satellite Systems (GNSS), and mobile data collection platforms such as QField played a central role in monitoring and analysing pest populations. The use of georeferenced data collection allowed precise recording of trap locations and damage observations throughout the vineyard. This information facilitated the creation of spatial distribution maps that revealed patterns of pest activity and damage occurrence. Such maps provide a visual representation of pest dynamics and allow managers to identify high-risk areas more effectively than conventional monitoring methods. Precision agriculture technologies also support the development of variable-rate management strategies. Instead of applying control

measures uniformly across the entire vineyard, interventions can be concentrated in areas where pest pressure is highest. This targeted approach reduces input costs, minimizes environmental impacts, and improves the efficiency of pest management programmes. The integration of remote sensing technologies, unmanned aerial vehicles (UAVs), and satellite imagery offers additional opportunities for vineyard monitoring. These technologies can provide information regarding vegetation health, canopy development, water stress, and other factors that influence pest susceptibility. When combined with trap monitoring data, remote sensing products contribute to a more comprehensive understanding of vineyard conditions and support predictive pest management strategies. Furthermore, the development of spatial databases and decision-support systems enables long-term analysis of pest trends and management outcomes. Historical monitoring data can be used to identify recurring infestation hotspots, evaluate the effectiveness of different control measures, and improve future management planning. The adoption of precision agriculture technologies therefore represents a significant advancement in sustainable vineyard management, enhancing both productivity and environmental stewardship.

11.5 Conclusion

The integration of pheromone trap monitoring, georeferenced damage assessment, and spatial analysis provided valuable insights into the dynamics of *Lobesia botrana* populations within the Quinta da Senhora da Graça vineyard. The results demonstrated a clear spatial relationship between trap captures and subsequent damage occurrence, confirming the effectiveness of pheromone traps as predictive tools for pest management. Terrain morphology was identified as an important factor influencing pest dispersion and the effectiveness of mating disruption strategies. Variations in topography, wind exposure, and microclimatic conditions contributed to the uneven distribution of captures and damage across the vineyard, emphasizing the importance of considering landscape characteristics in pest management planning. The study further demonstrated the value of integrated pest management strategies that combine preventive measures, biological control, monitoring, and selective chemical interventions. The mating disruption programme effectively reduced overall pest pressure, while monitoring activities enabled the identification of localized areas requiring additional attention. Finally, the application of precision agriculture technologies significantly enhanced the quality and usefulness of the monitoring programme. The use of GIS, GNSS, and digital field data collection systems facilitated detailed spatial analyses and supported evidence-based management decisions. These technologies provide powerful tools for improving pest control efficiency, reducing production costs, and promoting the long-term sustainability of vineyard production systems. Overall, the combination of spatial monitoring, integrated pest management, and precision agriculture technologies offers an effective framework for the sustainable management of *Lobesia botrana* and contributes to the production of high-quality grapes under environmentally responsible conditions.

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